Performance Modelling for Computer Systems

Endsem Solutions

Q1: Explain how you would simulate an M/M/K/K system on your computer. Write a pseudo code for it with necessary explanation

A:

a. What is M/M/K/K queue?: An M/M/K/K queue is a finite-capacity multi-server queue where:

- Arrivals follow a Poisson process with rate λ.
- Service times are exponentially distributed with rate μ.
- There are K servers available and the system can hold at most K customers (no waiting queue).
- Customers that arrive when all servers are busy are blocked (lost).

b. Simulation Explanation:

We will use discrete-event simulation to model the system. The events are:

(a) **Arrival Event:** A new customer arrives.
- If a server is free, the customer is assigned to it.
- If all K servers are busy, the customer is blocked.
- The next arrival is scheduled.

(b) **Departure Event:** A customer finishes service and departs.
- The corresponding server becomes free.

c. Pseudo Code Refer to algorithm 1

After simulation, we compute:

- **Blocking probability:**

$$P_{\text{block}} = \frac{\text{blocked customers}}{\text{total arrivals}} \tag{1}$$

- **Server utilization:**

$$U = \frac{\text{busy servers}}{K} \tag{2}$$

Q2: Consider an $M/M/\infty$ system. Model it as a Markov Chain and obtain its stationary distribution. Derive an expression for the mean number of jobs in the system and mean time spent by any job in the system.

A:

The Q matrix for the CTMC in Figure 1 is

$$\begin{bmatrix}
-\lambda & \lambda & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots \\
\mu & -(\mu+\lambda) & \lambda & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots \\
0 & 2\mu & -(2\mu+\lambda) & \lambda & \cdots & 0 & 0 & 0 & 0 & \cdots \\
\vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \cdots \\
0 & 0 & 0 & 0 & \cdots & k\mu & -(k\mu+\lambda) & \lambda & 0 & \cdots \\
0 & 0 & 0 & 0 & \cdots & 0 & (k+1)\mu & -((k+1)\mu+\lambda) & \lambda & \cdots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots
\end{bmatrix}$$

Algorithm 1 Simulation of an M/M/K/K System

```
Initialize K servers as free
Initialize total_arrivals = 0, blocked_customers = 0
Generate first arrival event
current_time ← 0
next_arrival_time ← current_time + Exp(λ)
Event Queue ← [(next_arrival_time, arrival)]
while simulation is running do
    Pop (event_time, event_type) from Event Queue
    current_time ← event_time
    if event_type == arrival then
        total_arrivals ← total_arrivals + 1
        if free servers > 0 then
            Assign customer to a free server
            Generate departure_time ← current_time + Exp(μ)
            Add (departure_time, departure) to Event Queue
        else
            blocked_customers ← blocked_customers + 1
        end if
        Schedule next arrival
        next_arrival_time ← current_time + Exp(λ)
        Add (next_arrival_time, arrival) to Event Queue
    else if event_type == departure then
        Free up the corresponding server
    end if
end while
```

Solving for $\pi Q = 0$

$$\begin{aligned}
\lambda\pi_0 &= \mu\pi_1 \\
(\lambda+\mu)\pi_1 &= \lambda\pi_0 + 2\mu\pi_2 \\
&\vdots \\
(\lambda+k\mu)\pi_k &= \lambda\pi_{k-1} + (k+1)\mu\pi_{k+1} \\
&\vdots
\end{aligned}$$

$$\implies \pi_k = \frac{1}{k!}\left(\frac{\lambda}{\mu}\right)^k \pi_0$$

Solving $\Sigma_{k=0}^{\infty}\pi_k = 1$

$$\begin{aligned}
\sum_{k=0}^{\infty}\pi_k &= 1 \\
\pi_0\left(\sum_{k=0}^{\infty}\frac{1}{k!}\left(\frac{\lambda}{\mu}\right)^k\right) &= 1 \\
\pi_0 e^{\lambda/\mu} &= 1 \\
\pi_0 &= e^{-\lambda/\mu}
\end{aligned}$$

Thus the stationary distribution is given by $\pi = [\pi_k]$ where

$$\pi_k = \frac{1}{k!}\left(\frac{\lambda}{\mu}\right)^k e^{-\lambda/\mu}$$

Let N denote the number of jobs in the system and T the time spent in the system then

$$\begin{aligned}
E[N] &= \sum_{k=0}^{\infty} k\pi_k \\
&= \sum_{k=0}^{\infty} k\frac{1}{k!}\left(\frac{\lambda}{\mu}\right)^k e^{-\lambda/\mu} \\
&= e^{-\lambda/\mu}\frac{\lambda}{\mu}\sum_{k=0}^{\infty}\frac{1}{(k-1)!}\left(\frac{\lambda}{\mu}\right)^{k-1} \\
&= \frac{\lambda}{\mu}
\end{aligned}$$

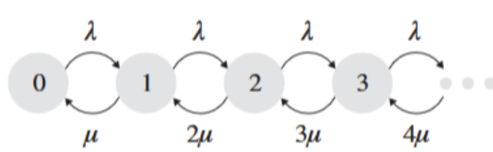

Figure 1: CTMC for $M/M/\infty$ system

Using Little's Law $E[N] = \lambda E[T]$

$$E[T] = \frac{1}{\mu}$$

Q3: Consider a system with two servers S_1 and S_2 with no buffer. Consider jobs arriving according to a Poisson process with rate λ. Every job must be served by first S_1 and then S_2. Server S_1 must be idle while S_2 is serving a job and therefore at a time only one job can be present in the system. Service time of a job at server S_i is arbitrary with mean μ_i. Use renewal reward theorem and time/ensemble averages to obtain the following: 1) probability that server S_i is busy for $i = 1, 2$. 2) Probability that system is completely idle with no jobs in the system.

A: From Theorem 3.7.1 (Sheldon Ross), we get:

$$\lim_{t\to\infty} P(X(t) = j) = \frac{E[\text{Amount of time in state } j]}{E[\text{Time of a cycle}]} \tag{3}$$

Here, a cycle C consists of:

- A : Arrival time,
- T_1 : Service time at S_1,
- T_2 : Service time at S_2.

Thus, the expected time of a cycle is:

$$E[C] = \frac{1}{\lambda} + \mu_1 + \mu_2 \tag{4}$$

The expected service times are:

$$E[T_1] = \mu_1, \quad E[T_2] = \mu_2 \tag{5}$$

Using renewal reward theorem, the probability that server S_1 is busy is given by:

$$P(S_1 \text{ busy}) = \frac{\mu_1}{\frac{1}{\lambda} + \mu_1 + \mu_2} \tag{6}$$

Similarly, the probability that server S_2 is busy is:

$$P(S_2 \text{ busy}) = \frac{\mu_2}{\frac{1}{\lambda} + \mu_1 + \mu_2} \tag{7}$$

The probability that the system is idle (i.e., no jobs present) is:

$$P(\text{Idle}) = \frac{\frac{1}{\lambda}}{\mu_1 + \mu_2 + \frac{1}{\lambda}} \tag{8}$$

Q4: For an $M/M/K/\infty$ system, derive the expression for 1) $E[N_q]$, the mean number of jobs waiting in the queue for their turn. (7 mks) 2) $E[W_q]$ the mean time spent by any job waiting in the queue before service. (3mks)

A:

The Q matrix for the CTMC in Figure 2 is

$$\begin{bmatrix} -\lambda & \lambda & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots \\ \mu & -(\mu+\lambda) & \lambda & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots \\ 0 & 2\mu & -(2\mu+\lambda) & \lambda & \cdots & 0 & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \cdots \\ 0 & 0 & 0 & 0 & \cdots & k\mu & -(k\mu+\lambda) & \lambda & 0 & \cdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & k\mu & -(k\mu+\lambda) & \lambda & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

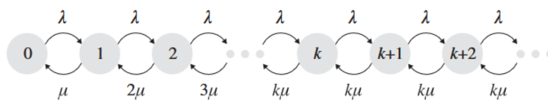

Figure 2: CTMC for $M/M/K$ system

Solving for $\pi Q = 0$

$$\begin{aligned}
\lambda\pi_0 &= \mu\pi_1 \\
(\lambda+\mu)\pi_1 &= \lambda\pi_0 + 2\mu\pi_2 \\
&\vdots \\
(\lambda+(k-1)\mu)\pi_{k-1} &= \lambda\pi_{k-2} + k\mu\pi_k \\
(\lambda+k\mu)\pi_k &= \lambda\pi_{k-1} + k\mu\pi_{k+1} \\
(\lambda+k\mu)\pi_{k+1} &= \lambda\pi_k + k\mu\pi_{k+2} \\
&\vdots
\end{aligned}$$

Solving gives us

$$\pi_i = \begin{cases} \frac{1}{i!}\left(\frac{\lambda}{\mu}\right)^i \pi_0 & \text{if } i \leq k \\ \frac{1}{k!}\frac{1}{k^{i-k}}\left(\frac{\lambda}{\mu}\right)^i \pi_0 & \text{if } i > k \end{cases}$$

Now N_Q is given by

$$\begin{aligned}
E[N_Q] &= \sum_{i=k}^{\infty}(i-k)\pi_i \\
&= \pi_0 \sum_{i=k}^{\infty}(i-k)\frac{1}{k!}\frac{1}{k^{i-k}}\left(\frac{\lambda}{\mu}\right)^i \\
&= \pi_0 \frac{k^k}{k!}\sum_{i=k}^{\infty}(i-k)\left(\frac{\lambda}{k\mu}\right)^i \\
&= \pi_0 \frac{k^k}{k!}\left(\frac{\lambda}{k\mu}\right)^k \sum_{i=0}^{\infty} i\left(\frac{\lambda}{k\mu}\right)^i \\
&= \pi_0 \frac{k^k}{k!}\left(\frac{\lambda}{k\mu}\right)^k \frac{k\lambda\mu}{(k\mu-\lambda)^2}
\end{aligned}$$

Using Little's Law $E[N_Q] = \lambda E[W_Q]$

$$E[T] = \pi_0 \frac{k^k}{k!} \left(\frac{\lambda}{k\mu}\right)^k \frac{k\mu}{(k\mu - \lambda)^2}$$

Q5: Let $A(t)$ and $D(t)$ denote two Poisson process with rates λ and μ respectively. Consider $N(t) = max(A(t) - D(t), 0)$. Prove/argue that $N(t)$ is a Markov chain. Identify its Q matrix and find its stationary distribution.

A: We define the process $N(t)$ as:

$$N(t) = \max(A(t) - D(t), 0)$$

where:

- $A(t)$ is a Poisson process with rate λ (arrivals).
- $D(t)$ is a Poisson process with rate μ (departures).

To verify that $N(t)$ is a Markov chain, we consider its transitions:

- $N(t) = n + 1$ occurs due to an arrival in $A(t)$.
- $N(t) = n - 1$ occurs due to a departure in $D(t)$, provided $N(t) > 0$.

Since the transitions depend only on the current state n and not on past states, $N(t)$ satisfies the Markov property and is a Continuous-Time Markov Chain (CTMC).

The process $N(t)$ represents the number of customers in an $M/M/1$ queue, where:

- Arrivals follow a Poisson process with rate λ.
- Service (departures) follows a Poisson process with rate μ.
- The system has a single server, and there is no limit on the queue length.

The generator matrix Q is given by:

- **Upper diagonal elements** $Q_{n,n+1} = \lambda$: These represent arrivals, which increase the $N(t)$ by one. Arrivals occur at a constant rate λ, independent of the queue length.
- **Lower diagonal elements** $Q_{n,n-1} = \mu$: These represent service completions (departures), which decrease $N(t)$ by one. Departures occur at rate μ, provided there is at least one customer in the system ($n > 0$).
- **Diagonal elements** $Q_{n,n}$: Each row sum in Q must be zero, so the diagonal elements represent the total rate of leaving a given state:

  $$Q_{n,n} = -(\lambda + \mu), \quad \text{for } n \geq 1.$$

  At $n = 0$, departures are not possible, so the departure rate μ is absent, and we have:

  $$Q_{0,0} = -\lambda.$$

$$Q = \begin{bmatrix} -\lambda & \lambda & 0 & 0 & 0 & \dots \\ \mu & -(\lambda+\mu) & \lambda & 0 & 0 & \dots \\ 0 & \mu & -(\lambda+\mu) & \lambda & 0 & \dots \\ 0 & 0 & \mu & -(\lambda+\mu) & \lambda & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

For the stationary distribution, the steady-state probabilities π_n satisfy the balance equations:
(Let $\rho = \frac{\lambda}{\mu}$)

$$\pi_{n+1} = \rho \pi_n$$

Using the normalization condition:

$$\sum_{n=0}^{\infty} \pi_n = 1$$

Substituting $\pi_n = \rho^n \pi_0$, we get:

$$\pi_0 \sum_{n=0}^{\infty} \rho^n = 1$$

Since the sum is a geometric series:

$$\sum_{n=0}^{\infty} \rho^n = \frac{1}{1-\rho}, \quad \text{for } \rho < 1$$

we obtain:

$$\pi_0 = 1 - \rho$$

Thus, the stationary distribution is:

$$\pi_n = (1-\rho)\rho^n, \quad \text{for } n \geq 0$$

where $\rho = \frac{\lambda}{\mu}$.